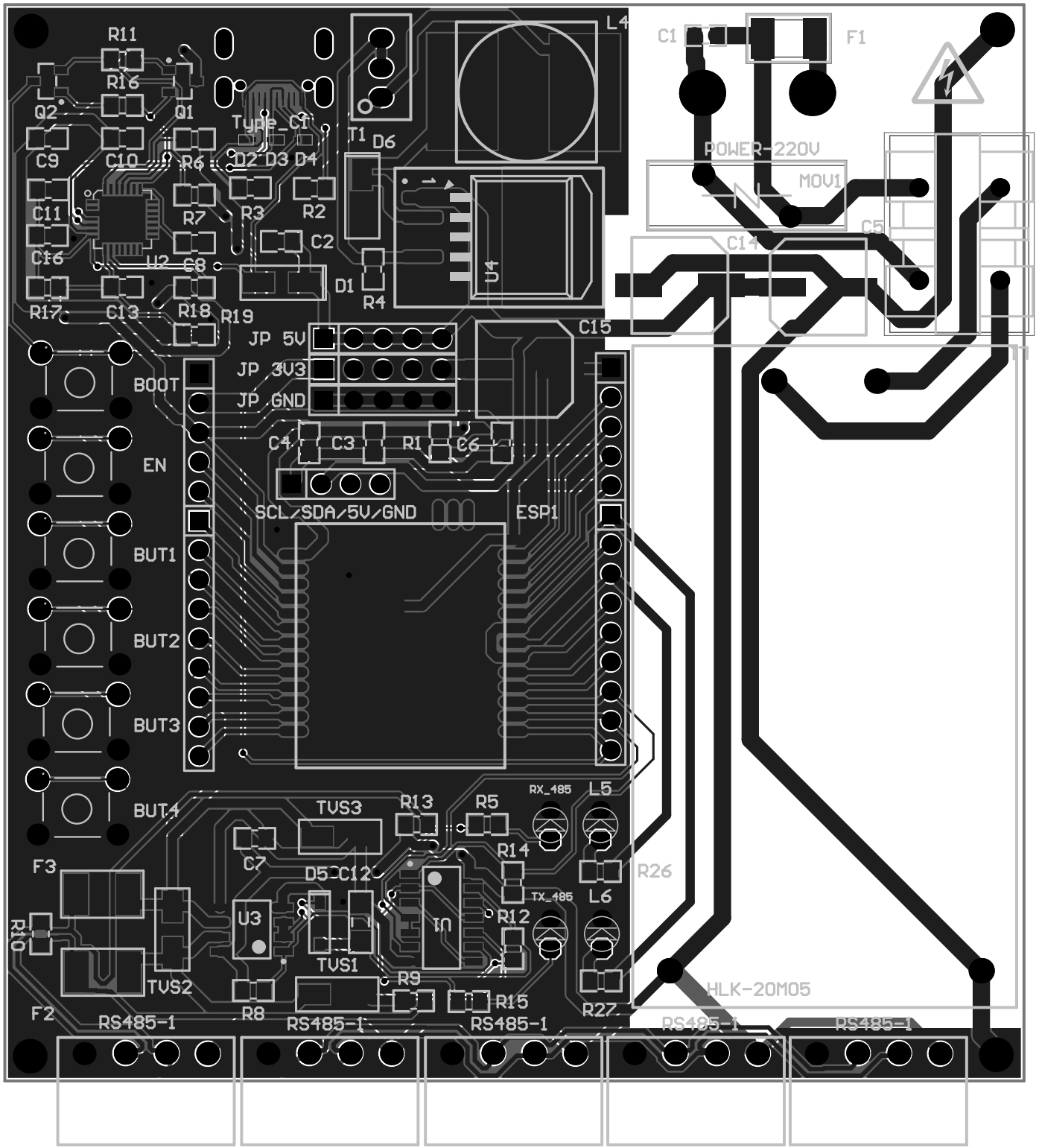

R11
R16
Q2
Q1
Type_C1
T1
D6
C9
C10
R6
D2
D3
D4
C11
R7
R3
R2
C2
C16
U2
C8
D1
R4
U4
R17
C13
R18
R19
JP 5V
JP 3V3
JP GND
BOOT
EN
BUT1
BUT2
BUT3
BUT4
C4
C3
R1
C6
SCL/SDA/5V/GND
ESP1
C15
L4
C1
F1
POWER-220V
MOV1
C5
C14
TVS3
R13
R5
RX_485
L5
C7
D5
C12
R14
R26
TX_485
L6
F3
U3
IN
R12
TVS1
R9
TVS2
R8
R15
R27
F2
HLK-20M05
RS485-1
RS485-1
RS485-1
RS485-1
RS485-1

Comment	Description	Designator	Footprint	LibRef	Quantity
BOOT	Nut Nhan 4 Chan 6x6mm	BT1	Nut Nhan 4 Chan Cam 6x6	Nut Nhan 4 Chan Cam 6x6	1
EN	Nut Nhan 4 Chan 6x6mm	BT2	Nut Nhan 4 Chan Cam 6x6	Nut Nhan 4 Chan Cam 6x6	1
BUT3	Nut Nhan 4 Chan 6x6mm	BT3	Nut Nhan 4 Chan Cam 6x6	Nut Nhan 4 Chan Cam 6x6	1
BUT1	Nut Nhan 4 Chan 6x6mm	BT4	Nut Nhan 4 Chan Cam 6x6	Nut Nhan 4 Chan Cam 6x6	1
BUT2	Nut Nhan 4 Chan 6x6mm	BT5	Nut Nhan 4 Chan Cam 6x6	Nut Nhan 4 Chan Cam 6x6	1
BUT4	Nut Nhan 4 Chan 6x6mm	BT6	Nut Nhan 4 Chan Cam 6x6	Nut Nhan 4 Chan Cam 6x6	1
104 0.1uF/275V	Dán	C1	tu 0805	Tu 0805 104 50V 10%	1
106	Dán	C2	tu 0805	Tu 0805 106 16V 10%	1
106 (10V/20%)	Dán	C3, C11, C13	tu 0805	Tu 0805 106 16V 10%	3
104 (10V/10%)	Dán	C4, C8, C9, C10, C16	tu 0805	Tu 0805 104 50V 10%	5
220uF	Dán	C5, C15	tu nhom 8 (e)	Tu Nhom 220uF 10V	2
105 (10V/10%)	Dán	C6, C17, C18, C19, C20, C21, C22	tu 0805	Tu 0805 105 50V 10%	7
104 100n	Dán	C7	tu 0805	Tu 0805 104 50V 10%	1
100pF	Dán	C12	tu 1206	Tu 1206 4.7nF 50V	1
100uF	Dán	C14	tu nhom 8 (f)	Tu Nhom 100uF 50V	1
1N5819(SS14)-SM	1.0A Schottky Barrier Rectifiers - Surface Mount	D1	SMA-J_DO-214AC	1N5819(SS14)-SM	1
LESD5D5.0CT1G		D2, D3, D4	LESD5D5.0CT1G,	LESD5D5.0CT1G	3
1N4148(T4)-SM	Fast Switching Diode - Surface Mount	D5	SOD123G_DO-215AD	1N4148(T4)-SM	1
1N5824(SS54)-SM	5.0A Schottky Barrier Rectifiers - Surface Mount	D6	SMA-J_DO-214AC	1N5824(SS54)-SM	1
ESP32-C5-WROOM-1/1U		ESP1	ESP32-C5-WROOM-1/1U	ESP32-C5-WROOM-1/1U	1
FUSE-CHIP-SM	Chip Fuses - Surface Mount	F1	FUSE-CHIP-1812-SM	FUSE-CHIP-SM	1
FUSE 1208	Chip Fuses - Surface Mount	F2, F3	FUSE-CHIP-1812-SM	FUSE-CHIP-SM	2
POWER-220V		HB1	HB9500	HB9500	1
HLK_20M05		HLK_1	HLK_20M05	HLK_20M05	1
JP esp	Cắm	JP1, JP2	JP 1x5 DT	JP 1x5 2.54 Duc Thang	2
JP esp	Cắm	JP3, JP4	JP 1x9 DT	JP 1x9 2.54 Duc Thang	2
SCL/SDA/5V/GND	Cắm	JP5	JP 1x4 DT	JP 1x4 2.54 Duc Thang	1
JP 5V	Cắm	JP6	JP 1x5 DT	JP 1x5 2.54 Duc Thang	1
JP 3V3	Cắm	JP7	JP 1x5 DT	JP 1x5 2.54 Duc Thang	1
JP GND	Cắm	JP8	JP 1x5 DT	JP 1x5 2.54 Duc Thang	1
10uH	EMI Filter Inductor - Through Hole	L1	L-UU9.8-L17.0-W12.0-H17.0-TH	L-FILTER-TH	1
RX_485	Cắm	L2	LED-3MM-RED	LED 3mm Do	1
TX_485	Cắm	L3	LED-3MM-RED	LED 3mm Do	1
33uH - 3A SMD	Dán	L4	Cuon Cam 12x12x10	CUON CAM 100uH SMD	1
LED_1	Cắm	L5	LED-3MM-RED	LED 3mm Do	1
LED_2	Cắm	L6	LED-3MM-RED	LED 3mm Do	1
10D561-560V	Metal Oxide Varistor - Through Hole	MOV1	MOV-P7.5-D17.0-T3.6-TH	MOV-TH	1
RS485-1	15EDG - 4P - 3.81 mm	P1, P2, P3, P4, P5	15EDG - 4P	15EDG - 4P	5
S8050(J3Y)-SM	0.5A 25V NPN General Purpose Transistors - Surface Mount	Q1, Q2	SOT23-3_TO-236AB	S8050(J3Y)-SM	2
10k	Dán	R1, R11, R16, R20, R21, R22, R23, R24, R25	tro smd 0805	Tro 0805 5% 10k	9
5.1k	Dán	R2, R3	tro smd 0805	Tro 0805 1% 5.1k	2
1.2k	Dán	R4	tro smd 0805	Tro 0805 5% 1.2k	1
1k	Dán	R5, R15, R18	tro smd 0805	Tro 0805 1% 1k	3
4.7k	Dán	R6, R7, R8, R9, R13, R14	tro smd 0805	Tro 0805 5% 4.7k	6
120R	Dán	R10	tro smd 0805	Tro 0805 5% 120R	1
56k	Dán	R12	TRO SMD 0805	Tro 0805 5% 56k	1
22k	Dán	R17, R19	tro smd 0805	Tro 0805 5% 22k	2
330R	Dán	R26, R27	Tro SMD 0805	Tro 0805 5% 330R	2
10K	Cắm	T1	Bien Tro Vi Chinh Dung	Bien Tro Tinh Chinh 10K Dung	1
SMAJ6.5CA-SM	ESD Suppressors/TVS Diodes 6.5V Bi-Dir - Surface Mount	TVS1, TVS2, TVS3	SMA-J_DO-214AC_TVS-BI	SMAJ6.5CA-SM	3
Type_C_16P		Type_C1	Type_C_16P	Type_C_16P	1
74HC14-SM	Hex Inverting Schmitt Trigger - Surface Mount	U1	SOIC14-1.27-150MIL	74HC14-SM	1
cp2102		U2	cp2102	cp2102	1
MAX485-SM	Low-Power, Slew-Rate-Limited RS-485/RS-422 Transceivers - Surface Mount	U3	SOIC8-1.27-150MIL	MAX485-SM	1
LM2976		U4	TO263	LM2596	1